Neuromorphic X2 Wishbone Pin Description

Hard-macro pin names, directions, bus requirements, and integration notes

Interface summary

The hard macro exposes a classic 32-bit Wishbone subordinate interface. Functional transfers use the byte address **0x3000_0004**, **wbs_sel_i = 4'b1111**, and **wbs_we_i = 1** for writes or **0** for reads. The Wishbone reset **wb_rst_i** is active high in the supplied behavioral model and functional wrapper.

The pin groups below expand to 128 physical pins and match the names represented in the LEF, Liberty, and GDS collateral. The supplied behavioral wrapper models the functional Wishbone path only; analog behavior and physical scan chains require separate signoff verification.

Wishbone and digital pins

Pin	Dir.	Width	Description
user_clk	IN	1	User-domain clock for hard-macro internal logic; unused by the functional behavioral wrapper.
user_rst	IN	1	Active-low user-domain reset for the physical macro; unused by the functional behavioral wrapper.
wb_clk_i	IN	1	Wishbone clock. Functional bus inputs are sampled on the rising edge.
wb_rst_i	IN	1	Active-high Wishbone and behavioral-model reset.
wbs_adr_i[31:0]	IN	32	Byte address. A functional transfer is accepted only at 0x3000_0004.
wbs_dat_i[31:0]	IN	32	Write data containing configuration or command words.
wbs_dat_o[31:0]	OUT	32	Read data containing results, status, or the empty-result sentinel 0xA000_0000.
wbs_ack_o	OUT	1	Acknowledges an accepted classic Wishbone transfer.
wbs_cyc_i	IN	1	Asserted by the initiator while a bus cycle is active.
wbs_stb_i	IN	1	Asserted with CYC to request a transfer.
wbs_we_i	IN	1	Write enable: 1 selects write; 0 selects read.
wbs_sel_i[3:0]	IN	4	Byte select. The model accepts only 4'b1111 full-word accesses.

Scan and test pins

Pin	Dir.	Description
ScanInCC	IN	Physical scan control input. Not functionally modeled by the hard-macro-compatible behavioral wrapper.
ScanInDL	IN	User-clock scan-chain data input. Not functionally modeled by the wrapper.
ScanInDR	IN	Wishbone-clock scan-chain data input. Not functionally modeled by the wrapper.
ScanOutCC	OUT	Physical scan-chain output. The functional wrapper drives a deterministic zero.
TM	IN	Physical macro test-mode input. Keep inactive for functional Wishbone operation.

Power, ground, and analog pins

Pin	Dir.	Nominal use
VSS	INOUT	0 V ground reference.
VDDA1, VDDA2	INOUT	1.8 V analog supply domains.
VDDC1, VDDC2	INOUT	1.8 V digital/core supply domains.
Vcc_read	IN	0.3 V read rail.
Vcc_set	IN	3.3 V array SET rail.
Vcc_wl_read	IN	0.7 V word-line read rail.
Vcc_wl_set	IN	1.8 V word-line SET rail.
Vcc_wl_reset	IN	2.6 V word-line RESET rail.
dc_bias	IN	External DC bias input; use the value defined by the signoff integration specification.
Iref	IN	100 uA low-noise reference current input.
Vbias	IN	1.8 V analog bias input.
Vcomp	IN	0.6 V comparator/reference bias input.
Bias_comp2	IN	0.6 V secondary comparator bias input.

Wishbone transfer requirements

Requirement	Customer integration rule
Address	Drive wbs_adr_i = 0x3000_0004 unless the behavioral ADDR_MATCH parameter is deliberately overridden.
Word access	Drive wbs_sel_i = 4'b1111. Partial-byte transfers are not acknowledged by the behavioral model.
Direction	Drive wbs_we_i = 1 for writes and 0 for reads.
Handshake	Hold CYC, STB, address, select, direction, and write data stable until ACK is observed. Deassert CYC and STB before the next classic transfer.
Write backpressure	A write can wait when the 32-word command FIFO is full.
Read behavior	A valid read is acknowledged immediately. If no result is queued, the normal read value is 0xA000_0000.
Reset	Assert wb_rst_i high to reset the behavioral Wishbone state. After reset, the next three accepted writes are configuration words.

Physical-collateral scope

LEF and GDS define physical geometry and pin placement. Liberty provides timing and integration data. The digital behavioral source provides functional Wishbone behavior. None of these artifacts alone replaces LVS, extracted timing, analog verification, or silicon characterization.